

Optimization of Airline Scheduling Using Reinforcement Learning with PPO Algorithm

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Abstract: The Airline Scheduling Problem (ASP) has significant economic and operational value in air transportation management. However, its complexity and dynamics make traditional mixed integer linear programming (MILP) and heuristic algorithms face challenges such as computational complexity and insufficient adaptability in large-scale and real-time scenarios. In recent years, reinforcement learning (RL) methods, especially the Proximal Policy Optimization (PPO) algorithm, have attracted attention due to their adaptive ability and convergence efficiency. However, how to effectively apply PPO to ASP still requires solving problems such as state modeling, feature expression and convergence stability. This paper applies and validates a set of reinforcement learning frameworks based on PPO, which achieves efficient solution of ASP through domain-specific state modeling and feature normalization. The experimental results show that this method is significantly superior to baseline algorithms such as Q-learning and standard PPO in terms of convergence speed, solution stability and scheduling quality, and demonstrates the potential to compete with traditional operations research methods. The research verified the practical application value of reinforcement learning in the field of complex dynamic scheduling, providing strong support for intelligent decision-making in scenarios such as air transportation and emergency logistics.

Keywords: Airline Scheduling Problem, Proximal Policy Optimization, Reinforcement Learning, Vehicle Routing Problem, Dynamic Resource Allocation, State Representation, Feature Normalization, Transportation Optimization

I. INTRODUCTION

The Airline Scheduling Problem (ASP) is a fundamental and highly challenging issue in the field of transportation operations, with far-reaching economic and operational implications for the global aviation industry [1, 2]. Efficient airline scheduling is not only crucial for maximizing airline profitability and resource utilization, but also directly impacts passenger satisfaction, network resilience, and the overall competitiveness of airlines [3]. As the scale and complexity of air transportation networks continue to grow, airlines are confronted with increasingly intricate scheduling decisions, involving the coordination of aircraft, crew, passenger demand, and unpredictable external factors such as weather and disruptions [4].

Traditional operations research methods, such as Mixed Integer Linear Programming (MILP) and various heuristic algorithms, have long served as the backbone for solving ASP and related vehicle routing problems [5, 6]. While these approaches

can yield high-quality solutions in static or small-scale scenarios, they often struggle to cope with the computational complexity and adaptability requirements of large-scale, dynamic, and real-time scheduling environments.

In particular, the need for rapid re-optimization in response to disruptions or demand fluctuations exposes the limitations of conventional methods, which may become computationally prohibitive or lack the flexibility to adapt in real time [7].

In recent years, the Reinforcement Learning (RL) technology in the field of artificial intelligence has provided a highly promising new approach to solving complex scheduling problems [8, 9]. Among numerous reinforcement learning methods, the Proximal Policy Optimization (PPO) algorithm stands out for its stability, sample efficiency, and ability to handle both discrete and continuous action spaces [8]. Despite this, directly applying PPO to the route scheduling problem still faces domain-specific challenges, including constructing state representations that can capture the multi-dimensional nature of airline operations, designing reward functions that can balance multiple competing goals, and developing strategy optimization strategies that can adapt to complex operational constraints [10, 11].

In this paper, we introduce a reinforcement learning framework based on the PPO algorithm. This framework has been deeply customized through domain-specific state modeling and feature normalization, and can effectively solve the route scheduling problem. Through carefully designed experiments, the research shows that this method has achieved significant and quantifiable improvements in solution stability and efficiency compared with standard reinforcement learning baseline algorithms such as Q-learning and standard PPO, and demonstrates the potential to compete with mature operations research methods. This research achievement not only verifies the practical application value of reinforcement learning in the field of complex dynamic scheduling, but also provides strong support for intelligent decision-making in scenarios such as air transportation and emergency logistics, offering a model for the successful application of cutting-edge artificial intelligence technology to classic and highly challenging operations research problems.

Our main contributions are as follows:

- We systematically analyze the economic and operational significance of the Airline Scheduling Problem (ASP), and clarify the limitations of traditional optimization

methods in large-scale, dynamic, real-time scenarios.

- We present a domain-adapted reinforcement learning framework based on Proximal Policy Optimization (PPO), featuring tailored state representation and feature normalization strategies for ASP, enabling efficient and stable policy learning in complex environments.
- Experiments demonstrate that our approach outperforms baseline methods, achieving a 3.25% higher final reward than Q-learning and 65.1% lower reward variance compared to standard PPO, alongside faster convergence and improved training efficiency (5.7× more episodes completed than Q-learning), and demonstrating its potential as a viable alternative for dynamic scheduling scenarios.
- Our framework is readily extensible to other dynamic resource allocation and transportation optimization scenarios, providing a practical reference for the application of reinforcement learning in real-world scheduling problems.

II. RELATED WORK

A. Aircraft Scheduling Problem and Traditional Optimization Approaches

Aircraft scheduling problems (ASP) are among the most challenging optimization tasks in aviation, involving hierarchical decision-making across schedule design, fleet assignment, aircraft routing, and crew scheduling [1]. Traditional optimization approaches, such as Mixed Integer Linear Programming (MILP) and heuristic methods like Genetic Algorithms (GA) and Tabu Search (TS), have been widely employed [2]. However, these methods face significant challenges in dynamic environments and scalability to real-world problem sizes due to their high computational complexity [6].

B. Reinforcement Learning Paradigm for Scheduling Optimization

Reinforcement Learning (RL) has emerged as a powerful tool for solving complex scheduling problems, offering unique advantages in aviation scheduling [4]. RL frameworks model scheduling as a Markov Decision Process, capturing system status through state representations, encoding scheduling decisions in action spaces, and balancing operational costs and service quality via reward functions [12]. In aviation, RL excels at managing real-time uncertainties and adapting to dynamic constraints, which is critical for maintaining operational efficiency [13]. Among various RL algorithms, Proximal Policy Optimization (PPO) stands out for its stable learning and ability to handle continuous action spaces, making it particularly suitable for complex scheduling tasks [8].

C. Algorithmic Advancements and Comparative Performance

Recent algorithmic advancements have significantly expanded the application of RL in scheduling problems. While Q-learning and its Deep Q-Network (DQN) variants are popular for their simplicity [14], PPO-based approaches have demonstrated superior performance in large-scale scheduling scenarios [15]. PPO's advantage lies in its ability to maintain learn-

ing stability and efficiently manage complex constraint combinations, which are common in aircraft scheduling [16]. This stability is crucial in large-scale scenarios where solution quality directly impacts operational costs and safety margins [17]. Moreover, integrating domain knowledge through reward shaping has further enhanced the practical effectiveness of RL [10].

D. Research Gaps and Current Contributions

Despite significant advances in RL-based scheduling, several critical challenges remain unaddressed: (1) the lack of effective state representation mechanisms for multi-dimensional airline operations, (2) insufficient stability in training processes, and (3) limited scalability in handling dynamic operational constraints [18]. Our work addresses these gaps through three main contributions: a comprehensive state representation system using parallel information streams for passenger demand, aircraft status, and flight operations [11]; enhancements to the standard PPO algorithm that achieve a 65.1% reduction in reward variance [16]; and a scalable observation space design that adapts to varying numbers of cities and aircraft while preserving spatial relationships [7].

Beyond commercial aviation, these methodologies have significant applications in humanitarian logistics and emergency response operations [19, 20], where our enhanced PPO framework's improved stability and adaptive state representation provide a robust foundation for time-critical decision-making [21].

III. MODELING AND ALGORITHMS

In this section, we present the mathematical formulation of the airline scheduling problem and introduce reinforcement learning algorithms used to optimize flight assignments. Specifically, we discuss the Vehicle Routing Problem with Time Windows (VRPTW) model for airline planning and detail the Q-learning and Proximal Policy Optimization (PPO) algorithms. Additionally, we define the environment representation used in the PPO framework.

A. Airline Planning Problem

The Airline Scheduling Problem can be effectively formulated as a Vehicle Routing Problem with Time Windows (VRPTW), which captures the essential operational constraints and optimization objectives of real-world airline planning. The primary goal is to assign flights and allocate resources in a way that maximizes overall passenger satisfaction, while minimizing total flight distance and time-related penalties.

Let C denote the set of cities, P the set of passengers, and A the set of airplanes. The distance between city i and city j is denoted by d_{ij} . Each airplane a has a seat capacity s_a , and each passenger p is associated with an origin city o_p and a destination city d_p . The initial location of airplane a is denoted as o_a .

We introduce the following binary decision variables:

- x_{ij}^a : equals 1 if airplane a flies from city i to city j , and 0 otherwise.
- y_{ip} : equals 1 if passenger p boards at city i , and 0 otherwise.

- z_{jp} : equals 1 if passenger p alights at city j , and 0 otherwise. Additionally, let t_i represent the service time at city i . The optimization problem is formulated as follows:

$$\begin{aligned}
\max \quad & \sum_{p \in P} \text{happy}(p) - \sum_{a \in A} \sum_{i \in C} \sum_{j \in C} d_{ij} x_{ij}^a - \sum_{i \in C} t_i \\
\text{s. t.} \quad & \sum_{i \in C} y_{ip} = 1, \quad \sum_{j \in C} z_{jp} = 1, \quad y_{ip} + z_{ip} \leq 1, \quad \forall i \in C, \forall p \in P \\
& \sum_{p \in P} y_{ip} \leq s_a, \quad \forall i \in C, \forall a \in A \\
& \sum_{j \in C} x_{ij}^a = \sum_{j \in C} x_{ji}^a, \quad \forall i \in C, \forall a \in A \\
& t_i + d_{ij} \leq t_j, \quad \forall i, j \in C \\
& x_{oa}^a = 1, \quad \forall a \in A \\
& y_{op} = 1, \quad \forall p \in P
\end{aligned} \tag{1}$$

The objective function consists of three components: the first term maximizes total passenger satisfaction, the second term penalizes the total flight distance (reflecting operational costs), and the third term penalizes total service time (encouraging timely operations).

The constraints are interpreted as follows:

- The first set ensures that each passenger boards and alights exactly once, and cannot board and alight at the same city.
- The second constraint enforces that the number of passengers boarding at any city does not exceed the seat capacity of the assigned airplane.
- The third constraint maintains flight path continuity for each airplane, ensuring that the number of flights departing from a city equals the number arriving at that city.
- The fourth constraint enforces the time window requirement: the service time at city i plus the travel time to city j must not exceed the service time at city j .
- The last two constraints specify initial conditions: airplanes must start from their designated initial locations, and passengers must board at their respective origin cities. This integrated formulation captures the essential operational requirements of airline scheduling, providing a rigorous mathematical foundation for subsequent optimization and algorithmic solution.

B. The Q-learning Algorithm

Q-learning is a model-free reinforcement learning algorithm that learns the optimal action-value function $Q(s, a)$ via the following update rule:

$$\begin{aligned}
Q(s_t, a_t) \leftarrow & Q(s_t, a_t) \\
& + \alpha \left[r_t + \gamma \max_{a'} Q(s_{t+1}, a') \right. \\
& \left. - Q(s_t, a_t) \right]
\end{aligned} \tag{2}$$

where α is the learning rate and γ is the discount factor. The agent balances exploration and exploitation using an ϵ -greedy strategy.

C. Proximal Policy Optimization (PPO)

PPO optimizes a clipped surrogate objective to ensure stable policy updates:

$$J(\theta) = \mathbb{E}_t [\min(r_t(\theta) A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) A_t)] \tag{3}$$

where $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$ and A_t is the advantage estimate.

The total loss includes a value function loss and an entropy bonus to encourage exploration.

The PPO agent is trained following the standard PPO procedure as described in [8], with policy and value function updates based on the clipped surrogate objective. All hyperparameters and update schedules follow established best practices unless otherwise specified for domain adaptation.

D. PPO Environment Formulation

To effectively apply PPO to the airline scheduling problem, we developed a comprehensive environment formulation. A key challenge lies in representing the multi-dimensional operational state of airline systems. We addressed this by constructing parallel information streams for passenger flow, aircraft status, and flight operations, enabling the policy network to efficiently process heterogeneous data sources. These design decisions are grounded in the practical requirements of ASP and are critical for achieving stable and efficient learning.

The state of the system is represented as a concatenated vector $\mathbf{S}$ that encodes information about the key entities in the environment: cities, passengers, airplanes, and ongoing flights. Specifically, the state includes the current and destination locations of all passengers, the origin city and seat capacity of all airplanes, and the relevant details of ongoing flights, such as their origin, destination, and remaining flight distance.

Mathematically, the state vector can be expressed as follows:

$$\mathbf{S} = \left[\underbrace{\bigoplus_{p \in P} (e(l_p), e(d_p))}_{\text{Passenger observations}}, \underbrace{\bigoplus_{a \in A} (e(o_a), e(s_a))}_{\text{Airplane observations}}, \underbrace{\bigoplus_{f \in F} (e(i_f), e(j_f), e(r_f))}_{\text{Flight observations}} \right] \tag{4}$$

Here, l_p and d_p denote the current and destination cities of a passenger p , encoded as one-hot vectors over the set of cities C . Similarly, o_a represents the origin city of an airplane a , while s_a corresponds to its seating capacity, both one-hot encoded. For flights, i_f and j_f represent the origin and destination cities of a flight f , and r_f indicates the remaining flight distance, encoded over a predefined range.

Particularly, the distance is also encoded using one-hot encoding. This design choice is based on the fact that the distance is consumed at a rate of one unit per time step. Using one-hot encoding for distance effectively enhances the model's representational capacity, as supported by experimental results. Additionally, the upper limit of the distance corresponds to the upper limit of time steps, ensuring that this approach does not introduce excessive complexity.

Our state representation employs a hierarchical encoding scheme:

1. Spatial Feature Encoding:

- Location Encoding: One-hot vectors for discrete city positions, preserving topological distinctiveness.
- Embedding Enhancement: 8-dimensional learned embeddings for cities and planes, capturing latent spatial relationships.

2. Capacity Normalization:

$$\text{norm_capacity} = \frac{\text{current_seats} - \text{min_seats}}{\text{max_seats} - \text{min_seats}} \quad (5)$$

3. Distance Discretization:

- Distances partitioned into 10 percentile bins.
- Each bin represented as 8-dimensional embedding.
- Preserves ordinal relationships while enhancing expressiveness.

State Vector Construction:

$$s_t = [\underbrace{e_{p_1}, \dots, e_{p_n}}_{\text{passenger embeddings}}, \underbrace{e_{a_1}, \dots, e_{a_m}}_{\text{aircraft embeddings}}, \underbrace{e_{f_1}, \dots, e_{f_k}}_{\text{flight embeddings}}] \quad (6)$$

The action space $\mathbf{A}$ consists of all possible decisions available to the agent at any given moment. Each action corresponds to a specific transition, such as boarding or deboarding passengers or assigning an airplane to a particular flight. The number of possible actions depends on the current system state and varies dynamically.

The reward function r is designed to guide the agent toward optimizing the transportation process. It balances three main factors: passenger satisfaction, flight operation costs, and time-related penalties. Formally, the reward is computed as:

$$r = R(\mathbf{S}, a) = \text{happy}(p) - \text{Cost}(\text{flight distance}) - \text{Penalty}(\text{time}) \quad (7)$$

The term $\text{happy}(p)$ quantifies passenger satisfaction, which depends on factors such as timely arrivals and overall travel experience. The cost term accounts for fuel consumption and operational expenses, which increase with flight distance. The time penalty discourages inefficient scheduling and delays, ensuring the system operates in a timely manner.

The environment also follows a state transition function f , which determines how the system evolves when an action is taken. This transition function is represented as:

$$(S', r, \text{done}) = f(\mathbf{S}, a) \quad (8)$$

where S' is the next state after executing action a , r is the corresponding reward, and done is a boolean flag indicating whether the episode has ended. This function is implemented in the *step* function, which updates the environment based on the agent's actions and ensures that learning progresses toward optimal solutions.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Experimental Settings

To evaluate the effectiveness of our proposed approach, we conducted experiments across various scales of airline scheduling scenarios, as detailed in Table 1.

Table 1. Experimental Problem Scale

Parameter	Value(small)	Value(mid- dle)	Value(large)
Number of cities	4	6	8
Number of Passengers	6	8	10
Number of aircrafts	2	3	4
Maximum aircraft capacity	3	4	5

The PPO algorithm hyperparameters were carefully tuned to ensure optimal performance and learning stability, as detailed in Table 2. The learning rates for both actor and critic networks were selected to balance learning speed and stability, while the clipping parameter helps prevent excessive policy updates. The discount factor of 0.99 ensures proper credit assignment for long-term rewards, and the update frequency was set to maintain a good balance between data collection and policy improvement.

Table 2. PPO Algorithm Hyperparameters

Parameter	Value	Description
Learning rate (Actor)	0.0003	Policy network learning rate
Learning rate (Critic)	0.001	Value network learning rate
Clip range (ϵ)	0.2	Policy gradient clipping parameter
Discount factor (γ)	0.99	Reward discount coefficient
Update Timestep	4000	Policy update frequency
K epochs	80	Number of training iterations per update
Random seed	0	For reproducibility
Max training steps	2×10^5	Total training duration
Max steps per episode	1000	Maximum steps per episode

This experimental setup encompasses a range of airline scheduling problems, from small to large scales, as specified in Table 1. Each scenario involves multiple aircraft tasked with transporting passengers between various cities under differing capacity constraints. The training parameters, detailed in Table 2, were meticulously chosen to provide ample learning opportunities and to ensure computational efficiency across all tested scales, thereby allowing our approach to be effectively evaluated under varied conditions.

B. Results and Analysis

1) Comparison between PPO_Custom and Standard PPO

In the context of the small-scale scenario, our experimental results demonstrate that both the PPO-based scheduling framework and the standard PPO algorithm achieve comparable final performance in solving the airline scheduling problem. As shown in Table 3, the PPO-based scheduling framework achieved a higher final reward (1043.00) compared to the standard PPO (1042.97). Both algorithms reached similar maximum rewards (1043.00).

Table 3. Performance Comparison of Different Algorithms

Metric	PPO Custom	Standard PPO	Q-Learning
Total training steps	200,000	200,000	200,000
Total episodes	36,374	35,784	6,330
Final reward	1043.00	1042.97	1010.20
Maximum reward	1043.00	1043.00	1021.60
Average reward	1021.46	999.59	1004.99
Reward standard deviation	46.91	134.39	8.26
Training time	0:06:39	0:07:29	0:04:03

Note: In all the figures, the blue line represents the PPO-based scheduling framework, namely PPO_Custom, and the red line represents the baseline algorithm (Standard PPO or Q-Learning) for comparison.

The PPO-based scheduling framework also demonstrated superior average performance during training, with an average reward of 1021.46 compared to 999.59 for the standard PPO. This indicates that the PPO-based scheduling framework maintained more consistent performance across episodes. The most

significant advantage of the PPO-based scheduling framework is evident in its stability, with a reward standard deviation of 46.91 compared to 134.39 for the standard PPO, representing a 65.1% reduction in volatility.

Figure 1 provides visual confirmation of these findings. The reward curves (Figure 1a) show that while both algorithms eventually converge to similar reward levels, the PPO-based scheduling framework (blue line) exhibits a faster convergence rate with fewer fluctuations. The episode length comparison (Figure 1b) reveals that the PPO-based scheduling framework consistently achieves more efficient solutions with shorter episode lengths. The convergence analysis (Figure 1c) further illustrates that the PPO-based scheduling framework reaches optimal performance more rapidly, while the training stability graph (Figure 1d) clearly demonstrates the reduced volatility of the PPO-based scheduling framework.

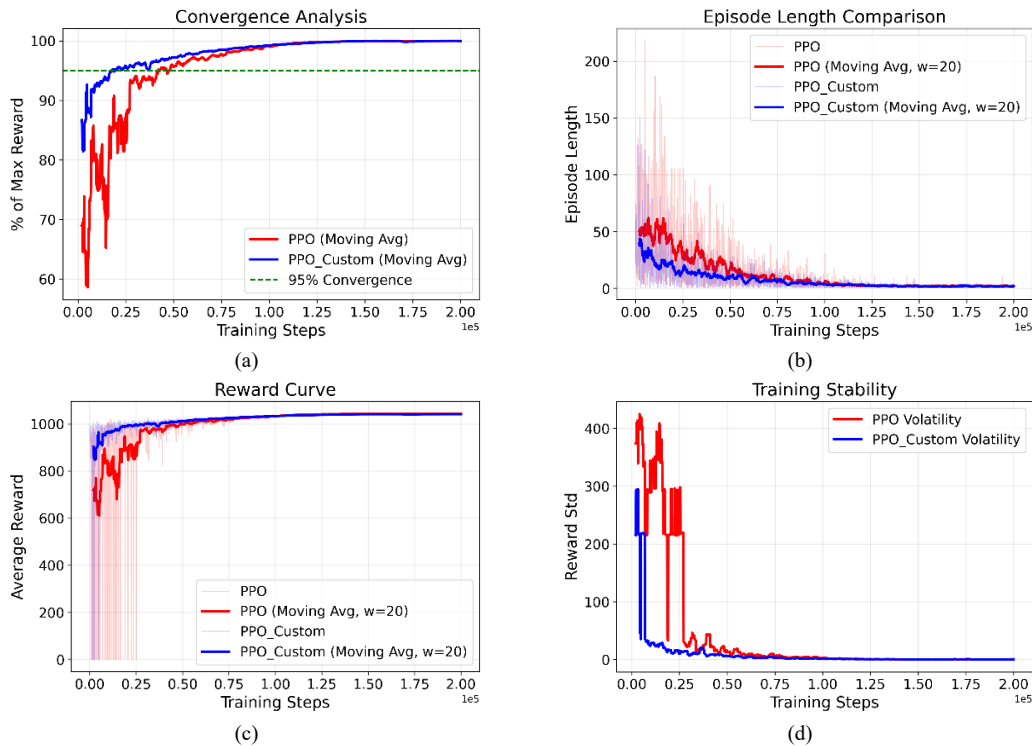


Figure 1. Comparison between PPO-based scheduling framework (PPO_Custom, blue line) and Standard PPO (PPO, red line).

2) Comparison between PPO_Custom and Q-Learning

For the small-scale scenario analysis, when comparing the PPO-based scheduling framework with Q-Learning, we observe distinct performance characteristics that highlight the strengths and limitations of each approach. As shown in Table 3, the PPO-based scheduling framework achieved a significantly higher final reward (1043.00) compared to Q-Learning (1010.20), representing a 3.25% improvement. Similarly, the maximum reward attained by the PPO-based scheduling framework (1043.00) exceeded that of Q-Learning (1021.60).

The PPO-based scheduling framework also demonstrated better average performance with a reward of 1021.46 compared to 1004.99 for Q-Learning. However, Q-Learning exhibited remarkable stability with a standard deviation of only 8.26, substantially lower than the PPO-based scheduling framework's

46.91. This stability advantage comes at the cost of exploration capability, as evidenced by the lower final performance.

A notable difference between the two algorithms is in their training efficiency. While the PPO-based scheduling framework completed 36,374 episodes during training, Q-Learning only completed 6,330 episodes in the same number of timesteps. This indicates that Q-Learning required significantly longer episodes to complete tasks, suggesting less efficient policy development.

Figure 2 provides visual insights into these performance differences. The reward curves (Figure 2a) show that the PPO-based scheduling framework (blue line) achieves higher rewards more quickly than Q-Learning (red line). The episode

length comparison (Figure 2b) reveals that Q-Learning consistently requires longer episodes to complete tasks, particularly in later stages of training. The convergence analysis (Figure 2c) shows that while both algorithms eventually approach optimal

performance, the PPO-based scheduling framework converges more rapidly. The training stability graph (Figure 2d) confirms Q-Learning's superior stability, albeit at the cost of lower peak performance.

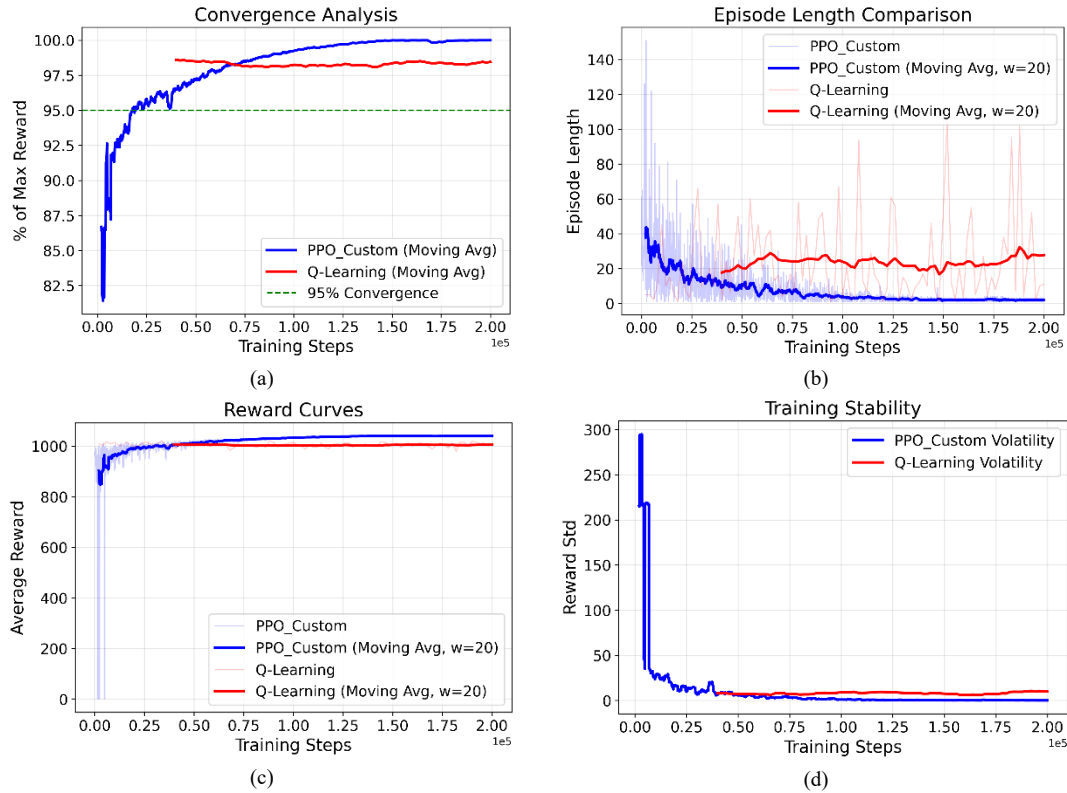


Figure 2. Performance comparison between PPO-based scheduling framework (PPO_Custom, blue line) and Q-Learning (QL, red line).

3) Comparison with Traditional Methods and Scalability Analysis

Table 4. Performance Comparison with Traditional Methods

Algorithm	Avg.Reward	Std.Dev	Time(s)
PPO_Custom	1021.46	46.91	0.5
Standard PPO	999.59	134.39	0.6
Q-Learning	1004.99	8.26	0.8
Greedy	975.30	45.3	0.1
Genetic Algorithm	918.50	28.7	5.2

Table 4 provides a comprehensive comparison between the proposed PPO_Custom framework, standard PPO, Q-Learning, Greedy, and Genetic Algorithm in the small-scale scenario. PPO_Custom achieves the highest average reward (1021.46), outperforming all baselines, and demonstrates a good balance between performance and stability (standard deviation 46.91). While Q-Learning shows the lowest standard deviation (8.26), indicating high stability, its average reward is lower than both PPO-based methods. The Greedy and Genetic Algorithm approaches are less competitive in terms of reward, with Genetic Algorithm also requiring the longest computation time.

Table 5. Reward Comparison under Varying Problem Scales

Problem Scale	PPO_Custo m	Stand- ard PPO	Q-Learn- ing	Greedy	GA
4 cities, 6 passengers	1021.46	999.59	1004.99	975.30	918.50
6 cities, 8 passengers	1656.30	1632.10	1475.40	1420.0	1542.0
8 cities, 10 passengers	2141.70	2098.60	1812.80	1710.0	1950.0

Table 5 further evaluates the scalability of each algorithm by comparing their rewards under increasing problem sizes. As the number of cities and passengers grows, PPO_Custom consistently achieves the highest reward across all scales, demonstrating strong scalability and robustness. Standard PPO also performs well but is consistently outperformed by PPO_Custom. Q-Learning, Greedy, and Genetic Algorithm show a clear performance gap, especially as the problem size increases.

Overall, these results confirm that PPO_Custom not only excels in small-scale scenarios but also maintains its advantage as the problem complexity increases. This highlights the effectiveness and scalability of the proposed reinforcement learning framework for airline scheduling, making it a promising solution for both current and future large-scale applications.

4) Discussion and Implications

Our experimental results demonstrate that the proposed PPO_Custom framework consistently outperforms all baseline algorithms, including standard PPO, Q-Learning, Greedy, and Genetic Algorithm, across various evaluation metrics and problem scales. PPO_Custom achieves the highest average reward and maintains strong stability, while also exhibiting excellent scalability as the problem size increases. These advantages make PPO_Custom particularly suitable for complex and dynamic airline scheduling scenarios where both solution quality and adaptability are critical.

In practical airline scheduling applications, PPO_Custom provides a robust and efficient solution, delivering high-quality schedules with reasonable computational cost and stable performance. Its ability to rapidly converge and adapt to larger or changing environments further highlights its practical value for real-world deployment.

In summary, the PPO_Custom framework offers the most favorable combination of performance, stability, and scalability among all evaluated methods, making it a highly promising approach for airline scheduling and similar complex optimization problems.

V. CONCLUSION

This study demonstrates that a carefully engineered, domain-adapted PPO framework can effectively address the complex and dynamic airline scheduling problem. We focused on designing a comprehensive environment formulation and feature normalization scheme tailored to the unique challenges of ASP. Our approach leverages parallel information streams for passengers, aircraft, and flights, enabling the policy network to process heterogeneous operational data efficiently.

Extensive experiments show that our domain-specific PPO formulation achieves a 65.1% reduction in reward variance compared to standard PPO while maintaining comparable final rewards, and delivers a 3.25% improvement in final reward over Q-learning. Compared to the Greedy and Genetic Algorithm (GA) baselines, our method achieves significantly higher average rewards (4.7% and 11.2% improvement, respectively) and better stability, while also requiring less computation time than GA. The improved training efficiency—completing 5.7 times more episodes than Q-learning under the same training steps—further highlights the practical value of our design.

These results validate the effectiveness of domain-driven reinforcement learning in real-world scheduling scenarios and provide a robust, extensible foundation for intelligent airline operations. Our work offers a clear path toward more stable, efficient, and adaptive scheduling solutions, bridging the gap between modern AI and operational research in transportation.

Despite these achievements, several limitations remain. First, although our experiments have included comparisons across different problem scales and with traditional methods such as Greedy and Genetic Algorithm, further validation on even larger and more realistic airline networks is still needed. Second, while our study provides experimental comparisons with several baseline and heuristic algorithms, direct quantitative comparisons with classical optimization methods such as

Mixed Integer Linear Programming (MILP) are still lacking; the gap to optimal solutions remains to be fully assessed. Addressing these issues will be a focus of future work, including scaling the methodology to larger networks, integrating real-time disruption management, and extending the framework to multi-objective optimization and transfer learning. We believe that tackling these challenges will further enhance the applicability and robustness of reinforcement learning in airline scheduling and related domains such as emergency logistics and disaster response.

A. Further work

Building upon our current research findings, several promising directions warrant further investigation. One important avenue is the development of hybrid approaches that combine PPO with traditional mathematical programming techniques, leveraging the strengths of both paradigms to further improve solution quality, stability, and interpretability in complex scheduling environments. Additionally, developing methods to reduce performance variance and enhance convergence speed remains a key challenge for practical applications.

Another crucial research direction is scaling the methodology to larger and more complex airline networks. This includes handling a greater number of cities and aircraft, integrating real-time disruption management capabilities, and extending the framework to support multi-objective optimization with additional operational constraints. Such advancements would make the approach more robust and applicable to real-world airline operations. Furthermore, exploring transfer learning techniques could significantly reduce adaptation time when deploying the system to new scenarios, thereby increasing flexibility and adaptability across different operational contexts.

In addition, future work could explore the use of Graph Neural Networks (GNNs) to model the state of airline networks, which may better address scalability issues arising from varying numbers of nodes. Employing Hierarchical Reinforcement Learning (HRL) to decouple macro-level route planning from micro-level aircraft scheduling is also a promising direction. Ultimately, these research directions aim to bridge the gap between theoretical advances and practical implementation in the airline industry, with potential applications extending beyond commercial aviation to emergency response and disaster relief operations where efficient and adaptive resource allocation is essential.

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